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PROTOTYPE MODEL FOR TRACKING INDOOR DEVICE USING BLUETOOTH

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ABSTRACT

Innovation and creativity by software engineers towards Bluetooth has really brought technologies one ahead closer to more development of the computer science and the future really are shaping out to be exciting .The trend of this era where people always are rushing out for making simple device more smarter, are always clinging for proper utilization of the application, which enhance the creativity of mind releasing the new ideas from the normal application which makes daily life more comfortable .The issue of missing and lost belongings is one that almost everyone has to deal with at some point in their life. It includes every age of groups from child to an adult person. Stating the fact, where most occurrences of lost items are within the household or within the indoor environment. The material of missing can be anything related to household. Tracking devices usually have been the more reliable, more helpful for finding the lost items, most flexible in the terms of accessing the misplaced items, but mentioning the fact of the era that, developers mostly focus in tracking the outdoor items through GPS such example are, your normal car key, your location when lost or tracking the criminal record through GPS. Being the comfortable platform but indeed it lacks for to purchase by most of users for to fit the cost in their pocket.

Keywords: Tracking Devices, Misplaced Items, Bluetooth.

I. INTRODUCTION

The increased use of wireless communication technology has led to design of different tracking device that integrate its use, especially in the smart phones as the most adept tracking platform with user friendly. Considering their role in contemporary life, where tracking devices have been the resource facilitators which have changed all impossible things as possible and hence have proved by creating such devices which are really so flexible for adapting or accessing even in tight compartment . In this regard, or by research, most of developers only focus in tracking the outdoor items through the means of GPS and stating the fact that most of the devices are costly in nature for to purchase by every users and adding the point that it works through the means of internet for its location [6]. This prototype is designed such that it completely focuses on tracking the indoor item, behaves as user friendly holding its properties where it can be able to track the indoor items without the mode of internet connection and holds the simple IOT (Bluetooth).

This Prototype model for tracking of devices using IOT enables users to handle their own items or things using their mobile devices. This prototype model is used for searching your lost, stolen, or missing indoor items .This prototype is designed to assist people with the hassle of losing items. This is tailored towards anyone and can be affordable in price for every user. Developing the platform for IOT(Bluetooth) not only makes users for to search their items , it also helps in every aspect that can be affordable to purchase ,can renew the battery life if errors found , can communicate without GPS OR GMS channel, can process in low consumption of power supply . This prototype model can operate in any environmental condition as it operates without internet.

Many people professional field or by individual users around the world, mostly deals with missing of their belongings within the indoors which probably leads to irritation for finding or memorizing the items that were kept earlier. User's demands for such platform for tracking the indoor items which can function even with the failure of internet connection can be affordable by every users pocket and especially the most important can be able to track the indoor items. The increased use of wireless communication technology has led to design of different tracking device that integrate its use, especially in the smart phones [1] as the most adept tracking platform with user friendly . Considering their role in contemporary life, where tracking devices have been the resource facilitators which have changed all impossible things as possible. The existing tracking devices which

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are flexible for adapting or accessing even in tight compartment. Living in this era, where the ideas of developers are resulting and shaping out their ideas as project and are focusing to make the simple devices smarter than it was, is benefiting and assuring users that any problem has its solutions. This prototype is designed such that it can be able to track the indoor items, by allowing the user to easily interact with the device, providing the facility for the results (output) by notification in android application and the sound from the device.

II. METHODOLOGY

Connection of the device can be viewed in Figure 1. The Arduino board is powered on by connecting it to a 9V power source (battery), once done; it becomes a source of power for the Bluetooth module HC-05. It works on serial communication. The android app is designed to send serial data to Arduino Bluetooth module when the button is pressed on the app. The Arduino Bluetooth module at other end receive the data and send it to the Arduino through the TX pin of the Bluetooth module connected to RX pin of Arduino. The code uploaded to the arduino check the receive data and compares it.

The process is six steps as follows:

1. Setup circuit HC-05 Bluetooth Module, ultrasonic sensor and Arduino Uno (VCC, Tx, Rx, GND, echoPin, trigPin)
2. Write Arduino code to control BUZZER and to measure distance using mobile app.
3. Create Mobile App using MIT App Inventor
4. Generate APK and send it to phone using MIT a12 Companion App
5. Pair HC-05 Bluetooth Module with mobile phone
6. Finally Test Mobile App, control buzzer and Measure distance connected to Arduino Uno.

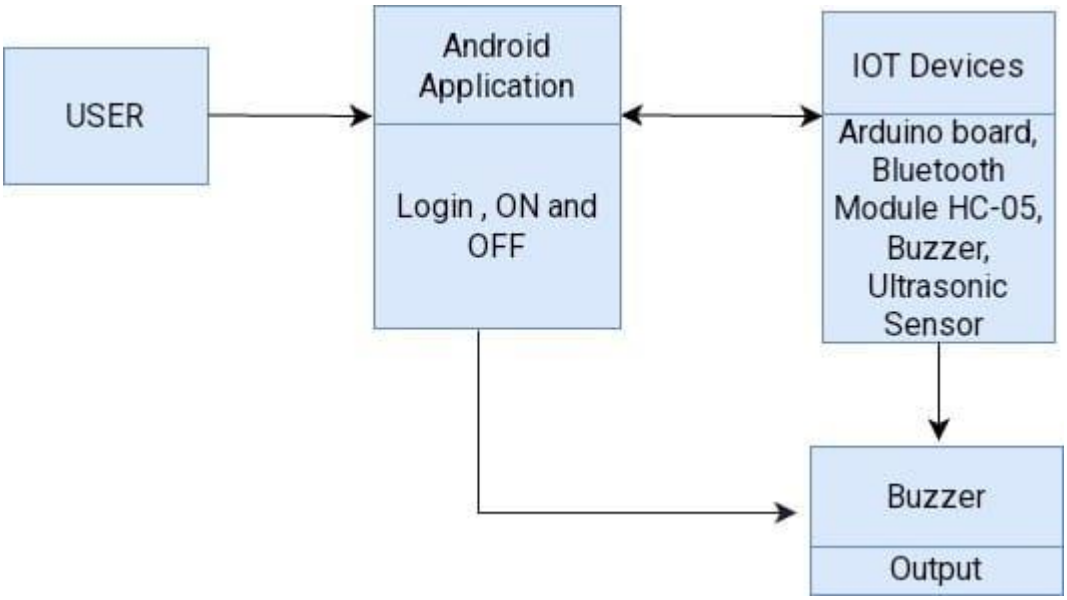

Figure 1: Methodology

III. RESULT

A. Results for Android Application

From (Fig 2 to Fig 6) it shows the Structure of the Android app. Fig 2 shows the Sign in page, which allows a user to gain access to an application by registering their username and password by any authentication user. Fig 3 shows the login page which allows a user to gain access to an application by choosing their username and password by any authentication user. In this page there is also a sign up button from where we can register the username and password. And Fig 4 shows the communication between the devices and application are shown in this code. This is the list picker of Bluetooth from where the device will communicate with software application. Once the button is turned on, the distance between the device and application is display in the software.

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Figure 2: Sign in Page

Figure 3: Log in Page

Figure 4: Username

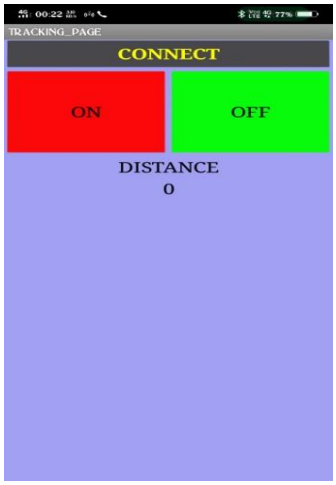

Figure 5: Tracking Page

Figure 6: Available Bluetooth

B. Results for Hardware Connection

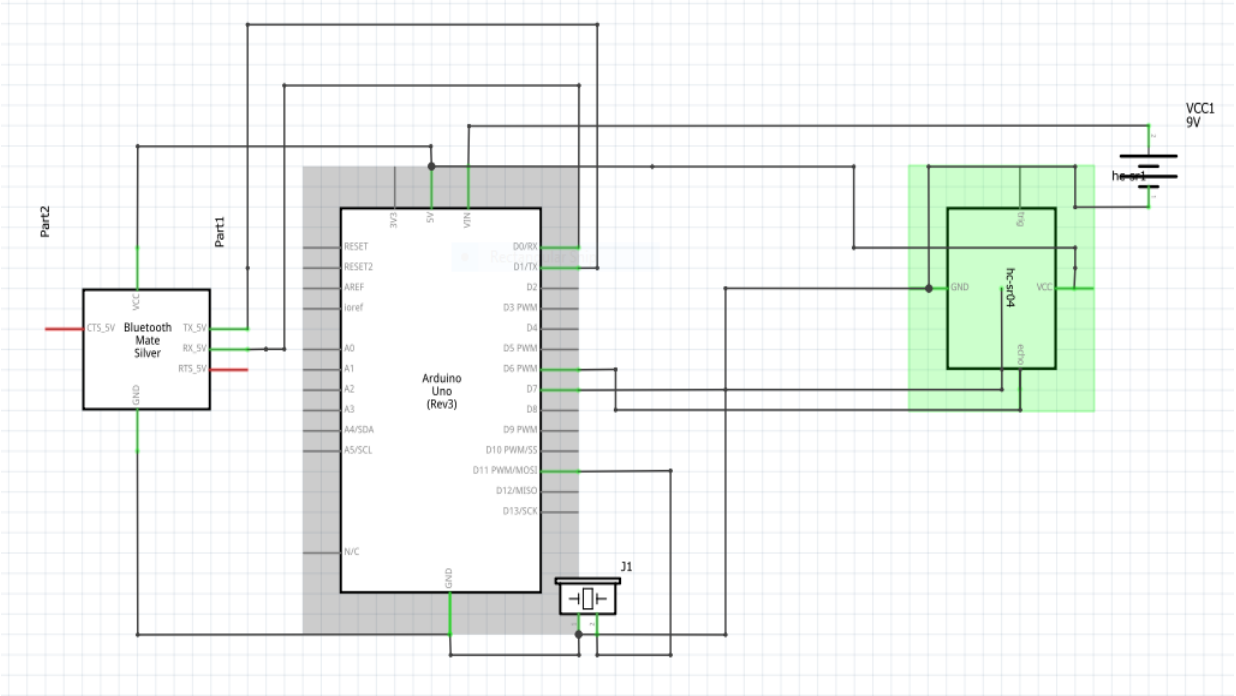

Figure 7: Circuit Diagram for Hardware Connection

As shown in Fig 7 Here's a textual representation of the Hardware Connections:

Arduino Uno Board:

- Connect the 5V pin of the Arduino Board to the power supply's positive terminal.
- Connect the GND pin of the Arduino Board to the power supply's negative terminal.
- Connect the all the necessary pins of the Arduino Board to the ultrasonic sensor, buzzer and Bluetooth module HC-05 as per the required connections.

Bluetooth Module HC-05:

- Connect the VCC pin of the Bluetooth Module HC-05 to the 5V pin of the Arduino Board.
- Connect the GND pin of the Bluetooth Module HC-05 to the GND pin of the Arduino Board.
- Connect the TX pin of the Bluetooth Module HC-05 to the RX pin of the Arduino Board.
- Connect the RX pin of the Bluetooth Module HC-05 to the TX pin of the Arduino Board.

Ultrasonic Sensor:

- Connect the VCC pin of the Ultrasonic sensor to the 5V pin of the Arduino Board.
- Connect the GND pin of the Ultrasonic sensor to the GND pin of the Arduino Board.
- Connect the Trig pin of the ultrasonic sensor to the Pin no.7 of the Arduino Board.
- Connect the echo pin of the ultrasonic sensor to the Pin no.6 of the Arduino Board.

Buzzer:

- Connect the positive terminal of the power supply to the 11 pin of the Arduino Uno board.
- Connect the negative terminal of the power supply to the GND pin of the Arduino Uno board

Power Supply:

- Connect the positive terminal of the power supply to the 5V pin of the Arduino Uno board.
- Connect the negative terminal of the power supply to the GND pin of the Arduino Uno board

IV. DISCUSSION

A. Study Limitation:

As we have successfully have accomplished our prototype model for tracking the indoor items. Since, the prototype model enable us to track the lost indoors items, where the users can detect their lost items through android application which is providing the distance of the lost items with sound .Keeping the developed prototype aside for a while, the fact that we somehow have failed in utilizing or implementing all of our ideas to this prototype model. In addition, we even feel at some point that if we have implemented all of ours ideas than it would really be enhancing the more features of the prototype and generating the interest of audience. Some ideas that we have failed to implementing our prototype model are as follows:

- **Proper distance:** The purpose of the prototype model was to determine the exact distance of the lost items. Since, our prototype model application provides you the distance of lost items .In this regard, observing the fact that if any obstacles (object) are placed in front of our prototype then it may lead to fluctuation of distance in application of the prototype.
- **Multiple access of lost items** : Most of people desires to have only one product for tracking their lost items which can access multiple items that are connected to the application, this creates efficiency of tracking the multiple items just by having one single product . Our developed prototype model is designed such that it can be able to track the indoor items but can just access only one item by one product.
- **Battery percentage :**Considering the role of battery percentage in day to day life where it acts as indicator to users to alert the percentage of battery remaining in your product , the purpose is to show or display the percentage that will be altering the validity of the battery that have been used . However, our prototype displays the voltage of the battery that have been used but at some point we do require of the battery percentage in the application

B. UPGRATION OF IDEAS

The prototype model for tracking devices using IOT (Bluetooth) was designed for just keeping the track of your lost items within indoors. As the model represents itself as prototype which is earlier version of a product from where the future version are developed. So we have mentioned all the components required by the prototype.

It really was a big challenge for connecting all hardware components inside Arduino board. So, as the hardware components increases, was so sure that it increases the size of product. As we used the hardware components for developing the prototype and hence our results were successful but at some point we cannot deny the fact that we couldn't find any solution for minimizing the product size.

- **Minimizing the Size:** Minimizing the size of product really would create a great impression in building the interest and attention of users to product. It will be enhancing the features of the prototype.
- **Application:** However, our product has an android application that holds its features such as it allows user for to easily interact with the application. As to state, the more modification the more benefit and the more demands of product.
- **Exact Distance:** The purpose of the prototype model was to determine the exact distance of the lost items. Since, our prototype model application provides you the distance of lost items .In this regard, observing the fact that if any obstacles (object) are placed in front of our prototype then it may lead to fluctuation of distance in application of the prototype.
- **Battery percentage :**Considering the role of battery percentage in day to day life where it acts as indicator to users to alert the percentage of battery remaining in your product , the purpose is to show or display the percentage that will be altering the validity of the battery that have been used . However, our prototype displays the voltage of the battery that has been used but at some point we do require of the battery percentage in the application.
- **Multiple access of items by one product:** Most of people desires to have only one product for tracking their lost items which can access multiple items that are connected to the application, this creates efficiency of tracking the multiple items just by having one single product .

V. CONCLUSION

The increased use of wireless communication technology has led to design of different tracking device that integrate its use, especially in the Smartphone's as the most adept tracking platform with user friendly. Considering their role in contemporary where tracking devices have been the resource facilitators which have changed all impossible thing as possible and have proved by creating such device which are really so flexible for adapting or accessing even in tight compartment. Innovation and creativity by software engineers towards Bluetooth has really brought technologies one ahead closer to more development of the computer science and the future really are shaping out to be exciting .The trend of this era where people always are rushing out for making simple device more smarter , are always clinging for proper utilization of the application, which enhance the creativity of mind releasing the new ideas from the normal application which makes daily life more comfortable. The issue of missing and lost belongings is one that almost everyone has to deal with at a some point in their live, it includes every age of groups from child to an adult person. Stating the fact, where most occurrence of lost items are within the household or within the indoor environment .The material of missing can be anything related to household such as your PDA'S. The behavior of human being who always choose the most beneficial and most easily accessible ways by which they can easily access their items without any means of trouble .Living in a tight compartment , tight schedule of daily life and a stress of an individual where most of the people tend to have common disease amnesia which causes fail to memorized the items that were kept earlier Our Prototype model for tracking indoor devices using Bluetooth enables users to handle their own items or things using their mobile devices. This prototype model is used for searching your lost, or missing indoor items .You can attach the tracking device to items and link into the app to track it from anywhere if you lose it. The app can also trigger a ringing or an alert noise from the tracking device to help locate your lost items, the Tracking devices not only provide detection of the items they also enhance the possibility of making users completely rely on these model for searching their misplaced items or unremembered stuff during a busy schedule. Our app is designed to assist people with the hassle of losing items. This app is tailored towards anyone and can be affordable in price for every users around the world without the means of internet, the user are freely flexible for to communicate without the usage of internet instead of internet it all uses is the Bluetooth for communicating the items that are linked inside your android application.

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